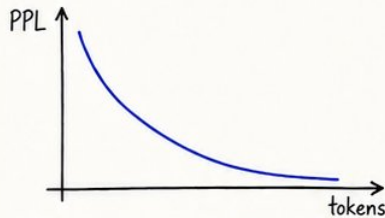


LLM Evaluation Metrics



Perplexity (PPL)



Measures how well the model predicts a sequence.

Lower is better.

$$\text{PPL} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(x_i) \right)$$

Accuracy (Exact Match)

Checks if the prediction exactly matches the reference.

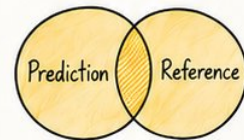


$$\text{EM} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i)$$

Higher is better.

F1 Score

Balances precision and recall for overlap-based evaluation.



$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Higher is better.

ROUGE-N / ROUGE-L

Measures overlap of n-grams or longest common subsequence.

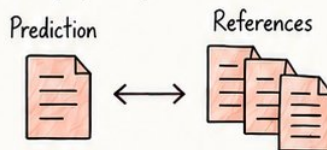


ROUGE-N, ROUGE-L

Higher is better.

BLEU Score

Evaluates n-gram overlap with brevity penalty.

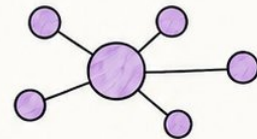


BLEU $\in [0, 1]$

Higher is better.

METEOR Score

Considers exact, stem, synonym matches and word order.

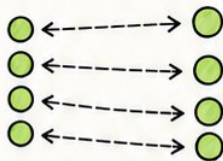


METEOR $\in [0, 1]$

Higher is better.

BERTScore

Uses contextual embeddings to measure semantic similarity.

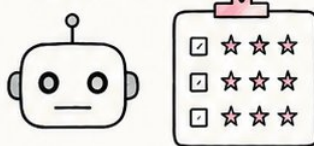


BERTScore $\in [0, 1]$

Higher is better.

GPT-4 as a Judge

Leverage LLMs to judge quality across criteria.



Score $\in [1, 5]$ or $[1, 10]$

Higher is better.

Human Evaluation

Human judgment on correctness, helpfulness, and safety.



Use rubrics & guidelines.

Higher is better.